METRIC

FF-B-171/2
30 NOVEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR, SINGLE ROW, RADIAL, NON-FILLING SLOT, ANSI/AFBMA DIMENSION SERIES 19.

The complete requirements for acquiring the bearings described herein shall consist of this document and the latest issue of FF-B-171/GEN.

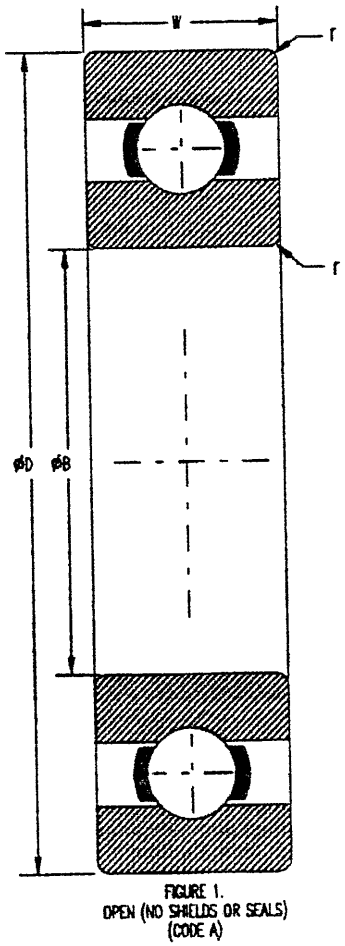


FIGURE 1.
OPEN (NO SHIELDS OR SEALS)
(CODE A)

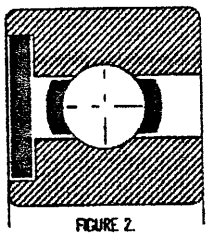


FIGURE 2.
SINGLE SHIELD
(CODE B)

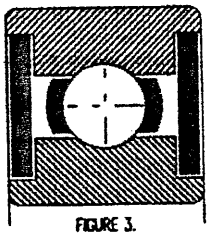


FIGURE 3.
DOUBLE SHIELD
(CODE C)

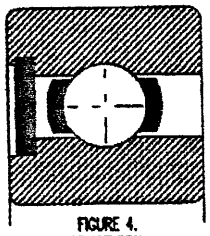


FIGURE 4.
SINGLE SEAL
(CODE D)

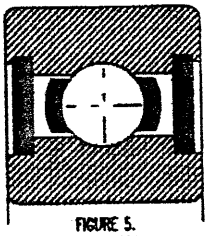


FIGURE 5.
DOUBLE SEAL
(CODE E)

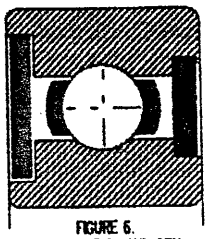


FIGURE 6.
SINGLE SHIELD AND SEAL
(CODE F)

BEARING CONFIGURATIONS

AMSC/NA

FSC 3110

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:45Z
Check the source to verify that this is the current version before use.

REQUIREMENTS :

1. Boundary dimensions. The boundary dimensions shall be specified in the part number by the size codes in Table I.

2. Cages. Cages shall be manufactured from any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

3. Shields and Seals. Shields and seals shall be fabricated from any material unless otherwise specified and shall be removable or nonremovable type. Seals shall be impervious contact type. Shields and seals shall be specified in the part number by the codes in table 111.

4. Radial Internal Clearance. The tolerance limits for radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

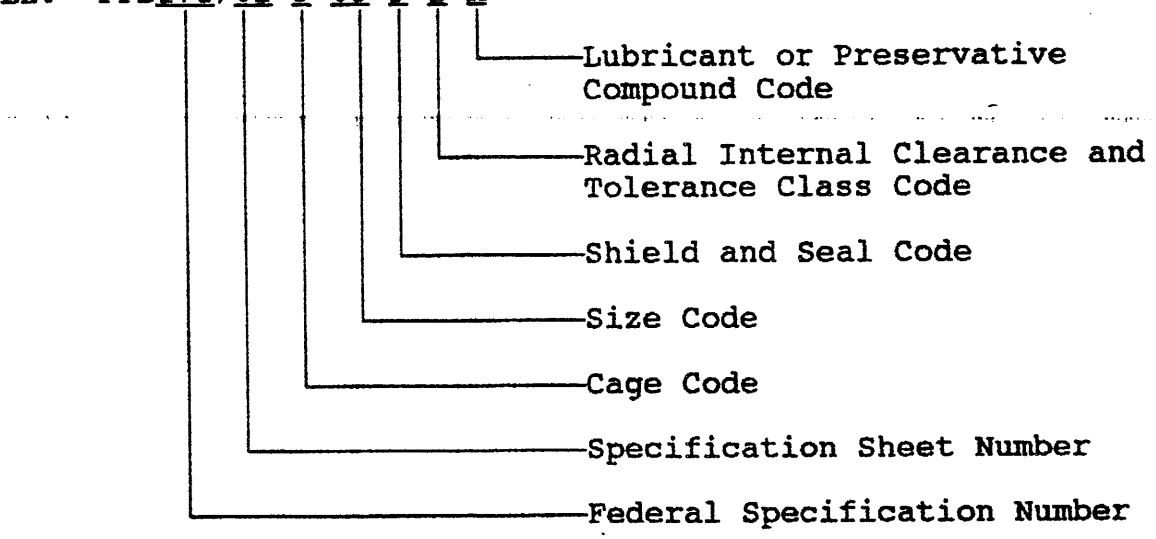
5. Tolerance Class. The tolerance limits for tolerance class ABEC-1 and ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

6. Lubricant or Preservative Compound. Preservative compound shall be applied to all open bearings (no shields or seals) and to single closure bearings (single shield or single seal) that have no additional lubricant requirement. Thus, for these bearings the bearing part number shall include the “C” suffix. For double closure or single closure bearings that require lubricant the bearing part number shall include the “A” or “B” suffix. When grease is required, the bearing void (the airspace between the inner and outer ring of the assembled bearing) shall be 25 to 50 percent filled. The lubricant or preservative compound shall be specified in the part number by the codes in table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:45Z
Check the source to verify that this is the current version before use.

7. Part number. The part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and alpha or numeric codes for cage, size, shields and seals, radial internal clearance and tolerance class, and lubricant or preservative compound.

EXAMPLE: FFB171/02 J 03 F 2 A



FFB171/02J03F2A indicates - Bearing, cage-pressed steel or molded plastic; bore 17mm; outside diameter 30mm; width 7mm; one shield and one seal; radial internal clearance symbol 0 (normal); tolerance class ABEC-1; grease in accordance with MIL-G-81322.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:45Z
Check the source to verify that this is the current version before use.

TABLE I. Boundary dimensions.

Size Code 1/	OB Bore	OD Outside Diameter	W Width	Chamfer $r_s$min 2/	Shaft Shoulder Diameter Min 3/	C Dynamic load rating (pounds)
00	10	22	6	0.3	12	475
01	12	24	6	0.3	14	530
02	15	28	7	0.3	17	735
03	17	30	7	0.3	19	795
04	20	37	9	0.3	22	1105
05	25	42	9	0.3	27	1215
06	30	47	9	0.3	32	1245
07	35	55	10	0.6	39	1645
08	40	62	12	0.6	44	2360
09	45	68	12	0.6	49	2430
10	50	72	12	0.6	54	2500
11	55	80	13	1.0	60	2860
12	60	85	13	1.0	65	2930
13	65	90	13	1.0	70	2990
14	70	100	16	1.0	75	4080
15	75	105	16	1.0	80	4190
16	80	110	16	1.0	85	4290
17	85	120	18	1.1	91.5	5490
18	90	125	18	1.1	96.5	5645
19	95	130	18	1.1	101.5	5795
20	100	140	20	1.1	106.5	7360
21	105	145	20	1.1	111.5	7295
22	110	150	20	1.1	116.5	7495
24	120	165	22	1.1	126.5	9460
26	130	180	24	1.5	138	11300
28	140	190	24	1.5	148	11500
30	150	210	28	2.0	159	14400
32	160	220	28	2.0	169	14700
34	170	230	28	2.0	179	14900
36	180	250	33	2.0	189	19000
38	190	260	33	2.0	199	19500
40	200	280	38	2.1	211	24800

1/ Size codes in Table I are industry bore codes.

2/ The minimum bearing chamfer ($r_s$min) must clear the maximum shaft or housing fillet radius. This dimension does not control the shape of the bearing chamfer contour.

3/ For reference purposes only. Maximum shoulder heights shall be such as to permit bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, provisions shall be made for the removal of the bearing from the shaft without damage to the bearing or other parts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:45Z
Check the source to verify that this is the current version before use.

TABLE II. Codes for Cages.

Code	Cage
J	Pressed steel or molded plastic
M	Machined bronze or brass
T	Machined non-metallic (phenolic)

TABLE III. Codes for Shields and Seals.

Code	Number of Shields and Seals.
A	none - fig.1
B	single shield - fig.2
C	double shield - fig.3
D	single seal - fig.4
E	double seal - fig.5
F	single shield and seal - fig.6

TABLE IV. Codes for Radial Internal Clearance and ABEC tolerance.

Code	Radial Internal Clearance	Tolerance Class
1	Symbol 2 (less than normal)	ABEC-1
2	Symbol 0 (normal)	
3	Symbol 3 (greater than normal)	
4	Symbol 4 (greater than symbol 3)	
5	Symbol 2 (less than normal)	ABEC-5
6	Symbol 0 (normal)	
7	Symbol 3 (greater than normal)	
8	Symbol 4 (greater than symbol 3)	

TABLE V. Codes for Lubricant and Preservative Compound.

Code	Lubricant or Preservative Compound
A	Grease in accordance with MIL-G-81322
B	Grease in accordance with DOD-G-24508
C	Preservative compound in accordance with MIL-P-197

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:45Z
Check the source to verify that this is the currentversion before use.